computer programmers. It is more interesting to classify strategies according to certain categories, and examine the success of these broader divisions. The most important category that Axelrod recognizes is ‘nice’. A nice strategy is defined as one that is never the first to defect. Tit for Tat is an example. It is capable of defecting, but it does so only in retaliation. Both Naive Prober and Remorseful Prober are nasty strategies because they sometimes defect, however rarely, when not provoked. Of the 15 strategies entered in the tournament, 8 were nice. Significantly, the 8 top-scoring strategies were the very same 8 nice strategies, the 7 nasties trailing well behind. Tit for Tat obtained an average of 504.5 points: 84 per cent of our benchmark of 600, and a good score. The other nice strategies scored only slightly less, with scores ranging from 83.4 per cent down to 78.6 per cent. There is a big gap between this score and the 66.8 per cent obtained by Graaskamp, the most successful of all the nasty strategies. It seems pretty convincing that nice guys do well in this game.

Another of Axelrod’s technical terms is ‘forgiving’. A forgiving strategy is one that, although it may retaliate, has a short memory. It is swift to overlook old misdeeds. Tit for Tat is a forgiving strategy. It raps a defector over the knuckles instantly but, after that, lets bygones be bygones. Chapter 10’s grudger is totally unforgiving. Its memory lasts the entire game. It never forgets a grudge against a player who has ever defected against it, even once. A strategy formally identical to Grudger was entered in Axelrod’s tournament under the name of Friedman, and it didn’t do particularly well. Of all the nice strategies (note that it is technically nice, although it is totally unforgiving), grudger/Friedman did next to worst. The reason unforgiving strategies don’t do very well is that they can’t break out of runs of mutual recrimination, even when their opponent is ‘remorseful’.

It is possible to be even more forgiving than Tit for Tat. Tit for Two Tats allows its opponents two defections in a row before it eventually retaliates. This might seem excessively saintly and magnanimous. Nevertheless Axelrod worked out that, if only somebody had submitted Tit for Two Tats, it would have won the tournament. This is because it is so good at avoiding runs of mutual recrimination.

So, we have identified two characteristics of winning strategies: niceness and forgivingness. This almost utopian-sounding conclusion—that niceness and forgivingness pay—came as a surprise to many of the